

# Radchaneeporn Changpun

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[Portfolio](#) | [LinkedIn](#) | [GitHub](#) | [Google Scholar](#)

## PROFESSIONAL SUMMARY

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AI and data science professional with an M.Sc. in Computer Science (Chulalongkorn University, GPA 3.94/4.00), focused on designing and building AI agents, LLM/RAG systems, and applied machine learning. Strong in Python and modern agent frameworks (LangGraph, LangChain, LlamaIndex, AutoGen) across the full experiment-to-evaluation lifecycle, with hands-on experience supporting deployment alongside production teams. Skilled at translating business requirements into solution architecture, integrating AI with APIs and data, and communicating clearly with technical and non-technical stakeholders in international teams. Published AI researcher and AWS Certified AI Practitioner.

## EDUCATION

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### M.Sc. in Computer Science, Chulalongkorn University

2023 – 2025

*Faculty of Engineering (GPA 3.94 /4.00)*

**Relevant coursework:** Artificial Intelligence, NLP, Pattern Recognition, Data Science & Data Engineering Tools, Data Science Architecture, Computer Security, Research Methods in CS

**Publication:** Changpun, R. et al., "Agentic Stage-Based LLM Framework for Multi-Turn Mental Health Support Conversations in Thai," ISAI-NLP 2025, IEEE. [Paper](#)

### B.Sc. in Chemical Engineering, Chulalongkorn University

2017 – 2021

*Faculty of Engineering*

**Publication:** Nimmanterdwong, P.; Changpun, R. et al., "Applied Artificial Neural Network for Hydrogen Sulfide Solubility in Natural Gas Purification," ACS Omega, 2021. [Paper](#)

## WORKING EXPERIENCE

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### PulseMetrics Pte Ltd

Oct 2025 – Present

*Senior Associate, Data Science*

- Work with CRM team client across two business units (Retail; Automotive Fast Fix), framing business questions and turning analytical results into clear, actionable insights for data-driven decisions
- Design and optimize high-throughput preprocessing pipelines for large-scale telematics sensor data (Big Data), applying data engineering principles to ensure data quality and reduce latency for downstream analytics and modeling

### AIMET, Center of Excellence in Digital and AI for Mental Health

Jul 2024 – Aug 2025

*Researcher*

- Designed and built an end-to-end multi-agent LLM system (stage-based, approach-selection, and monitoring agents) for Thai mental-health support, with human-in-the-loop controls and stage-transition governance
- Owned the research lifecycle: ran experiments across multiple approaches and built AI evaluation and feedback loops; the system reached a 79% positive user-reaction rate and improved 7 of 8 support metrics in practitioner testing
- Worked closely with the deployment team to move the system into production, gaining hands-on experience with deployment, monitoring, and MLOps practices
- Applied Responsible AI and domain-alignment practices in a sensitive, regulated domain, collaborating with psychology experts
- Led and mentored a cross-functional team of researchers, developers, and interns through design and implementation

### Wood PLC (Foster Wheeler)

Jan 2022 – Jan 2023

*Chemical Process Engineer*

- Designed and optimized chemical and energy processes for Thai oil & gas refinery clients at an international engineering consultancy, using data-driven analysis to improve operational efficiency

## SELECTED PROJECTS

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### LLM-Based Thai Personal Income Tax Q&A Chatbot [\[GitHub\]](#)

- Ran an empirical architecture evaluation across four LLM patterns (Agentic RAG, Naive RAG, Long-Context LLM, Vanilla Inference) to select the best approach for a regulated domain
- Assessed trade-offs with NLP metrics (BLEU, ROUGE-L, BERTScore) and LLM-as-a-judge, demonstrating evidence-based architecture decisions
- Built the data pipeline with OCR and web scraping; deployed an interactive UI with Gradio

### RAG Pipeline with Open-Source LLMs (Llama 2-13B) [\[GitHub\]](#)

- Architected a RAG pipeline connecting an LLM to an enterprise knowledge base (Scopus) via a Pinecone vector database with embedding-based semantic grounding
- Defined the RAG solution pattern (prompt orchestration, vector search, grounding) to reduce hallucinations; integrated with LangChain

### Multi-Label Text Classification with RoBERTa [\[GitHub\]](#)

- Fine-tuned RoBERTa on Scopus data for a 40.3% Macro-F1 improvement (0.6687 vs 0.1894 baseline) via transfer learning and a custom architecture, using PyTorch and Hugging Face

### Radchaneeporn Lab: Technical Content & Documentation [\[YouTube\]](#) [\[GitHub\]](#)

- Publish tutorials on modern LLM frameworks and document strategies for LLM cost optimization and system reliability

## SKILLS

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- **AI Agents & LLM Engineering:** Multi-agent system design, agent orchestration, RAG & Agentic RAG, prompt engineering & orchestration, vector search, semantic embeddings, long-context inference, fine-tuning (LoRA, QLoRA), human-in-the-loop controls
- **Machine Learning & Statistics:** Regression, classification, clustering, decision trees, neural networks, transformers; statistical & mathematical modeling; experiment design & evaluation, feature engineering
- **Evaluation & Engineering Practice:** AI evaluation frameworks, LLMOps concepts, AI observability, Responsible AI, debugging, system design, and data handling; hands-on exposure to model deployment and monitoring alongside deployment teams
- **Cloud & Data:** AWS (Certified AI Practitioner), cloud-based model training; Big Data processing; data-pipeline design; data quality; Pinecone vector database
- **APIs, Databases & Integration:** REST APIs, data-platform integration
- **Languages, Frameworks & Tools:** Python, Pandas, NumPy, Matplotlib, PyTorch, Scikit-Learn, Hugging Face, LangGraph, LangChain, LlamaIndex, AutoGen, Gradio, Weights & Biases (WandB), Git, Linux, Tableau, Next.js, TypeScript, PostgreSQL
- **Collaboration & Communication:** Cross-functional leadership, mentoring, stakeholder alignment, international team experience
- **Industry Domains:** Retail, Energy (Oil & Gas, Natural Gas processing), Healthcare (Mental Health), Automotive Fast Fix

## CERTIFICATION

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AWS Certified AI Practitioner, Amazon Web Services

Mar 2026